

Trees :

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16:25 hrs.

A tree is a hierarchical data structure composed of nodes and without any closed path (cycle).

Root - The top most node, or the starting point of the tree is called root. It is the first node of the tree and there is only one root for every non-empty tree.

Parent - The node connected directly above the current node is called the parent node of the current node. Every node (except root) has one parent node.

Child - The node connected below the current node is called the child of the current node.

Leaf - A node that has no children is called leaf.

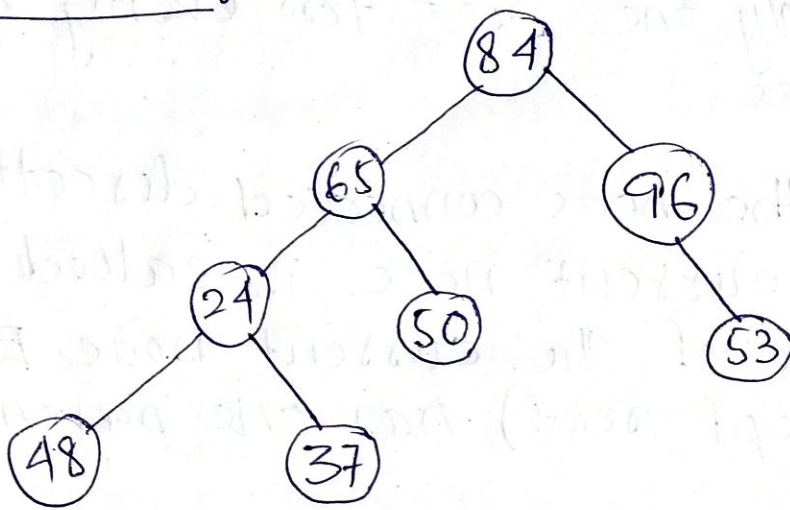
Depth/Level - The length of the path (edges) from the ~~tree~~ root to the current node is called the depth or level of the current node. Root node has depth 0.

Tree height - It is the maximum depth of any node.

Binary tree - It is a tree having nodes with at most two childrens.

It is commonly used in computing like data compression, file storage etc.

Example:



This is a binary tree with
root = 84

leaves = 48, 37, 50, 53

height = 3

depth of the node 50 is = 2

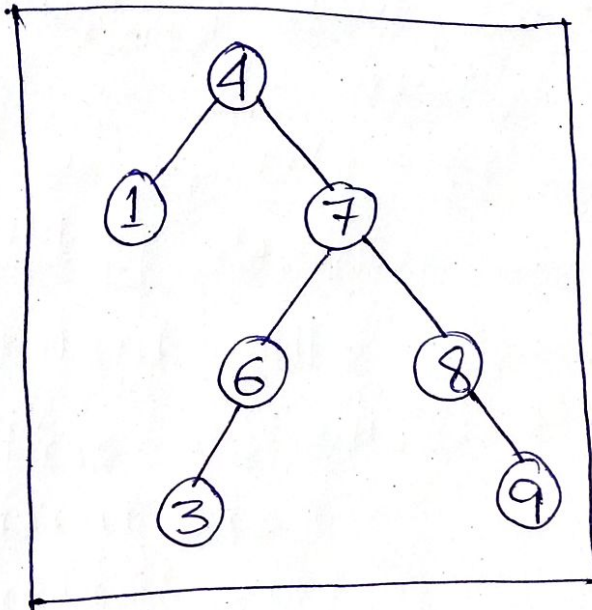
Binary search tree - A binary search tree (BST) is a binary tree with no duplicate nodes and follows the below ordering invariant principles-

⊙ At any node with value K ,

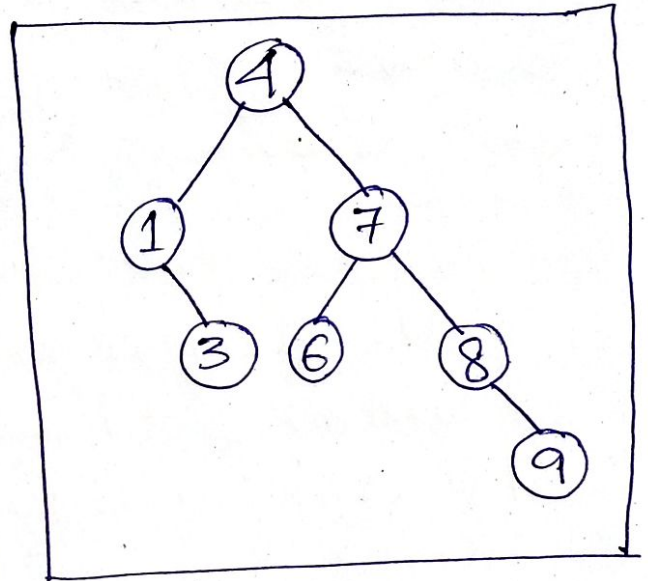
i) all values of nodes in the left side are strictly less than k .

ii) all values of nodes in the right side are strictly greater than k .

example: Is this a BST?



No



Yes

Draw BST : 84, 41, 96, 24, 37, 50, 13, 98.

